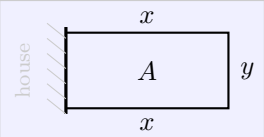


## L2.6a Optimization

§3.7 — optimization equation, constraint, interval

### The optimization template

| Step                      | Sarah's fence — 40 ft of fencing, the house is one side, and neither side may be under 5 ft                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ① Picture                 |                                                     |
| ② Sort the question       | <b>OEQ</b> $A = xy$<br><b>Constraint</b> $2x + y = 40$ , so $y = 40 - 2x$<br><b>Interval</b> $x \in [5, 17.5]$                       |
| ③ Find critical pts       | $A = x(40 - 2x) = 40x - 2x^2$<br>$A' = 40 - 4x = 0$ , so $x = 10$<br>back into the constraint: $2(10) + y = 40$ , so $y = 20$        |
| ④ Does this Maximize/min? | $A'' = -4$ , so $A$ is CD everywhere, so $x = 10$ is a Max<br><i>unsure which it is? that is what the 2nd Derivative Test is for</i> |
| ⑤ Check endpoints         | $A(5) = 150$ $A(10) = 200$ $A(17.5) = 87.5$                                                                                          |

*The question tells you what to optimize. The constraint tells you what to eliminate.*

### ELIMINATION WITH THE CONSTRAINT

An optimization equation in two variables cannot be differentiated. Use the constraint to write one variable in terms of the other, substitute, and *then* take the derivative.

$$A = xy \quad \text{with} \quad y = 40 - 2x \quad \implies \quad A(x) = 40x - 2x^2$$

*“if you find yourself differentiating the constraint, you have swapped the two”*

## THE INTERVAL SLOT

The interval is the third slot, and it is the one the picture does not always hand you. It comes from **both** sources at once — what the geometry allows, and what the question restricts.

**from the geometry** a length cannot be negative, so  $x \geq 0$

**from the question** “at least 5 ft wide/long” gives  $x \geq 5$  and  $y \geq 5$

The second one binds at *both* ends.  $x \geq 5$  reads off directly; and  $y \geq 5$  with  $2x + y = 40$  gives  $2x + 5 \leq 40$ , so  $x \leq 17.5$ .

$$x \in [5, 17.5]$$

*“not every interval is  $[0, \text{something}]$  — read the restrictions”*

## ENDPOINT CHECK

A critical number is a candidate, not an answer. Evaluate the optimization equation at **every** critical number *and* at **both** ends, then compare the list.

$$A(5) = 150$$

left end

$$A(10) = \mathbf{200}$$

critical number

$$A(17.5) = \mathbf{87.5}$$

right end

**Max**

**min**

The Max is interior, where the derivative put it. **The min is at the right endpoint** — and the derivative never proposed it. That is what step ⑤ is for.

Then **back-substitute**:  $x = 10$  is not what was asked for. The question wanted the dimensions, so finish —  $x = 10$  ft,  $y = 20$  ft, largest area  $200 \text{ ft}^2$  @  $x = 10$ .

*“an endpoint answer is not a mistake; it is often the answer”*

## WATCH OUT

~~X the constraint is the thing you maximize~~ → the OEq is what you optimize; the constraint is what lets you eliminate

~~X  $A = xy$ , so  $A' = x'y + xy'$~~  → eliminate first. One variable, then differentiate — nothing here is a function of  $t$

~~X the critical number is the answer~~ → it is a candidate. Step ④ or step ⑤ still has to say which kind it is

~~X the interval is always  $[0, \text{something}]$~~  → read the restrictions. Sarah's is  $[5, 17.5]$ , and the min lives at the right end

~~X an open interval means you skip the justification~~ → it means there are no ends to check, so the justification is the whole argument — use the 2nd Derivative Test

~~X  $x$  is the answer~~ → back-substitute. The question asked for the dimensions, the cost, or the time